



MSc Games Engineering: Dissertation Workshop

Kurt Debattista



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Dissertation

- ▶ Major individual project
- ▶ Develop skills
 - Research
 - Academic
 - Analytical



Dissertation

- ▶ Specific area of Games Engineering
 - Further develop career and personal interests
- ▶ 60 CATs
 - 600 hours
- ▶ Max word count: 15,000 words





Dissertation Process

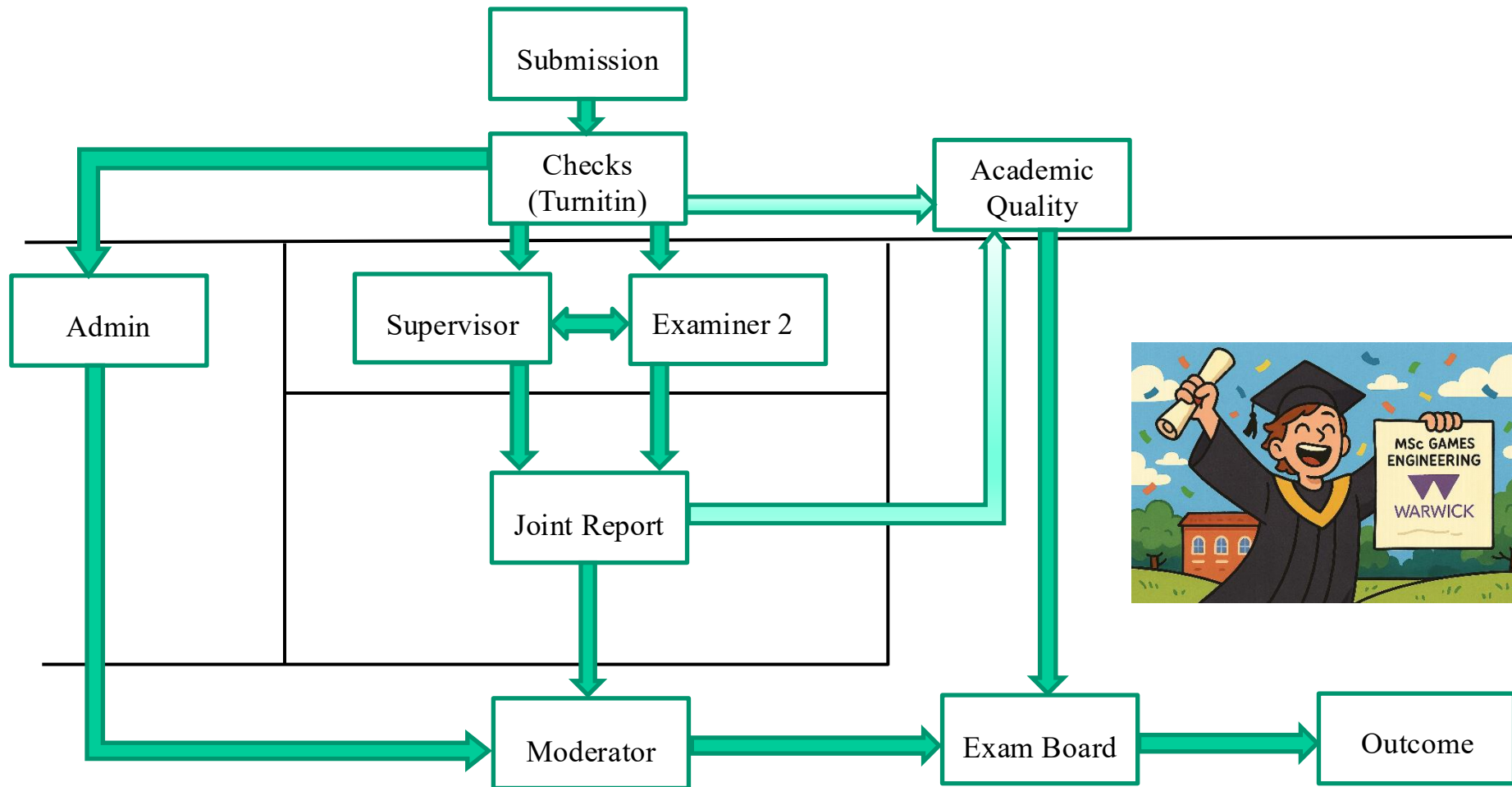
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Dissertation

- ▶ Submit on tabula
 - 12:00 noon on some day early September
- ▶ **Submit to your supervisors**
 - well in advance of this date
 - allow time for them to review your proposal
 - allow time for you make any amendments

<https://warwick.ac.uk/fac/sci/wmg/intranet/student/deptguidelines/academic/ftprojects/>





Outcomes

- ▶ Reviewers reach consensus on mark
- ▶ $\geq 80\%$ Outstanding quality
- ▶ $\geq 70\%$ Distinction
- ▶ $\geq 60\%$ Merit
- ▶ $\geq 50\%$ Pass
- ▶ $< 50\%$ Fail





Dissertation Content

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Contents of Dissertation

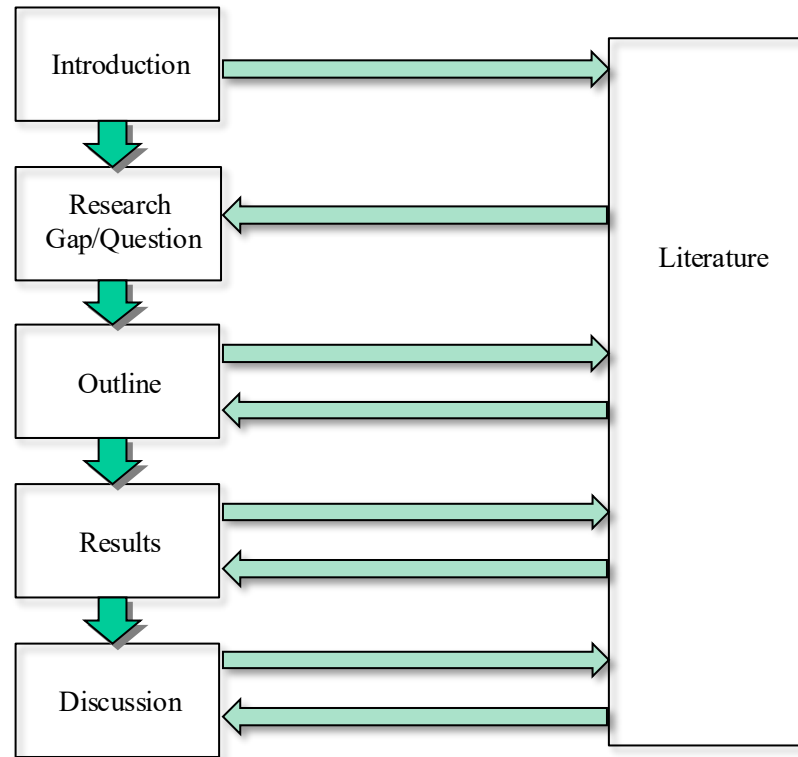
- ▶ Abstract
- 1. Introduction
- 2. Literature Review
- 3. Research Methodology/Outline
- 4. Method
- 5. Results and Analysis
- 6. Discussion
- 7. Conclusions
- ▶ Appendices



Literature Review

- ▶ Background
- ▶ Related Work
- ▶ Research Gap
 - Critical analysis
 - What is missing





Being Critical - Research Gap

	inplace?	stable?	worst	average	best	remarks
selection	✓		$N^2 / 2$	$N^2 / 2$	$N^2 / 2$	N exchanges
insertion	✓	✓	$N^2 / 2$	$N^2 / 4$	N	use for small N or partially ordered
shell	✓		?	?	N	tight code, subquadratic
merge		✓	$N \lg N$	$N \lg N$	$N \lg N$	$N \log N$ guarantee, stable
quick	✓		$N^2 / 2$	$2 N \ln N$	$N \lg N$	$N \log N$ probabilistic guarantee fastest in practice
3-way quick	✓		$N^2 / 2$	$2 N \ln N$	N	improves quicksort in presence of duplicate keys
???	✓	✓	$N \lg N$	$N \lg N$	N	holy sorting grail

Image Courtesy of Sedgewick and Wayne "Algorithms, 4th Edition slides"

Research Methodology/Outline

- ▶ What is the rationale for your research?
- ▶ What is your research question/problem/aim?
 - Objectives
- ▶ Overall experimental design
 - What methods and techniques will you use and why?
- ▶ Where/why will your data come from?
 - How will you access the data?
- ▶ How are results measured and why?
- ▶ How are you going to analyse the data?



Research Methodology/Outline

- ▶ Research aim/question
- ▶ Research objectives
 - Outline
 - Some detail
- ▶ Measure of success
- ▶ Data



Method

- ▶ Technical details
- ▶ Algorithms
- ▶ Equations/formulae
- ▶ Software engineering solutions
- ▶ Experimental design



Results and Analysis

- ▶ Outputs from experiments
- ▶ Comparisons with other methods
- ▶ Analysis of these results
- ▶ Tables
- ▶ Graphs



Discussion

- ▶ Not necessarily required
 - Can fold into Analysis or Conclusions
- ▶ Potential impact
- ▶ Opportunities & wider benefits
- ▶ Limitations



Abstract

- ▶ Concise overview of the document
- ▶ Rationale for the research
- ▶ Specific problem being addressed
- ▶ Outcome of literature review/research gap
- ▶ Proposed approach
- ▶ Current Progress
- ▶ Next stages
- ▶ Conclusions at this stage



Introduction

- ▶ Background
- ▶ Context
- ▶ Motivation and rationale
- ▶ Suitable for non experts (to an extent)
- ▶ Enough detail to demonstrate expertise
- ▶ Research aim/problem/question
- ▶ Overview of dissertation



Conclusions

- ▶ Contributions
- ▶ Work conducted
- ▶ Limitations (if not in discussion)
- ▶ Future work



Appendices

- ▶ Extra information
 - Keep brief





Exercise



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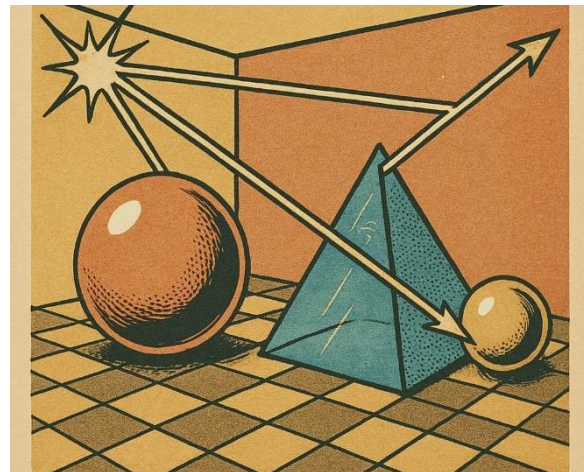
Exercise

- ▶ Optimising CPU-based Ray Tracing for Real-Time Interactive Applications
- ▶ Consider what goes into each chapter



Motivation

- ▶ Memory
- ▶ Flexibility
 - Enable quick changes
 - Test new models
 - Complexity and branching
- ▶ Other modalities
 - Physics, acoustics
- ▶ Off-load tasks
 - Subdivision structures, path guidance



Introduction

► Context

- Importance of ray tracing
 - Applications
- Challenges of ray tracing
 - Challenges on CPU
- Why CPU?



Introduction

- ▶ Identify RQ
 - What data structures and algorithms are most effective for accelerating ray-triangle intersection on the CPU?
 - How can we leverage parallelism to improve CPU ray-tracing performance?
 - What are the benefits of constructing spatial acceleration structures on the CPU for GPU-based ray tracing in dynamic scenes?



Introduction

- ▶ Specification (eg)
 - focusing on primary rays vs. Whitted vs. GI
 - specific scene complexities
 - specific CPU architectures
- ▶ Dissertation structure



Literature Review

- ▶ Background
 - Ray tracing vs rasterization
 - CPU vs GPU
- ▶ Related work
 - CPU ray tracing / GPU ray tracing
- ▶ Gap
 - What is missing from existing works



Research Methodology/Outline

- ▶ Research Question
 - Make clear
- ▶ Objectives
 - Lit. review
 - SIMD and MT ray tracing
 - Concurrent data structure creation and update
 - Results

Research Methodology/Outline

- ▶ Software tools
 - Raw C++, Unreal, Unity
 - Other software
- ▶ Test scenes
 - Vary complexity
 - Static/animated



Research Methodology/Outline

► Performance metrics

- Computation
- Quality (psnr, vdp)
- Comparisons (vs. ?)
- Other metrics
 - Rays/second
 - Speedup
 - Power consumption
 - Human factors



Method

- ▶ Details of algorithm
 - SIMD and MT
 - Concurrent data structure
- ▶ Implementation details
- ▶ Software architecture



Results and Analysis

- ▶ Graphs and tables for metrics
 - Per scene, per rays cast, per thread
 - Compared with other methods
- ▶ Explain results and identify patterns
- ▶ Qualitative results
 - Images, videos



Discussion

- ▶ What do results mean in terms of RQ?
- ▶ Is it feasible?
 - How does it compare against other methods?
 - GPU?
 - Performs poorly vs GPU but
 - more power efficient for mobile use
 - Easier to deploy on multiple architectures
 - When to use?
- ▶ Limitations



Conclusions

- ▶ Summarise
- ▶ Contributions
- ▶ Future work
 - Implement over multiple computers





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Turnitin Example

Upgrade Proposal and Turnitin

Early drafts

- ▶ It is your responsibility to ask your Supervisors, or the RDO, to upload the **first few pages** you write to Turnitin and discuss the report with you

The Final Proposal

- ▶ The final proposal will be uploaded to Turnitin
 - the report sent to Supervisors and Reviewers
- ▶ If plagiarism is identified
 - the examination is stopped and a formal investigation initiated



AI

▶ [AI Explainer](#)

▶ [University guidelines](#)





Academic Writing

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Clear and precise language

- Be very explicit with stating the contribution
 - “The main contributions of this work are ...”
 - Give bullet points

Formal tone

- Not too complex
 - Avoid “we” if possible (depends on field; sometimes appropriate)

Objectivity

- Evidence-based claims
- Use citations



Logical flow and structure

Signpost

- “In the next chapter” / “in Section ...”

Use figures and tables

- Figures can also be useful for showing methods and their detail

Define

- Acronyms
- Technical terms
- Terms in an equation



“We looked at the data and saw that things were different between the two groups. This shows our method works pretty well and it’s probably better than some other ones.”

“We analysed the XY dataset (XY, 2024) comprising 2,000 samples and observed a statistically significant difference ($p < 0.01$) between the control and treatment groups. These results suggest that our proposed method demonstrates improved accuracy compared to existing baseline approaches (Smith et al., 2024, Bianchi et al., 2025).”

“We ran a number of simulations to see how the system behaved. We saw improvements on a number of occasions but not always; which might have been because of the randomness. In the overall, we believe that the new approach is better than the old one as most of the graphs show better performance.”

“To evaluate the performance of the proposed control system, we conducted 100 Monte Carlo simulations under varying initial conditions and noise profiles, see Table 5. The system consistently achieved stability in 93% of trials, with failure cases linked primarily to outlier configurations exceeding the design assumptions, see Figure 7. Compared to the baseline controller (Bond, 2007), our method demonstrated a 17% improvement in average settling time and a 12% reduction in steady-state error. While some variance was observed in edge-case scenarios, the overall results support the robustness of our approach under practical conditions.”



Conclusions

- ▶ Follow structure
- ▶ Start writing early
 - Provide early drafts to supervisor
- ▶ Listen to your supervisor





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